as a result of the straightening process shall be borne by the Contractor.

**421.47 Misfits –** For all primary connections, and secondary connections having eight or more bolts, 85% of the holes shall accept bolts without reaming. The remaining 15% may be reamed to accept the designed diameter bolts. The diameter of the reamer shall be the same as the drilled hole.

Holes in plates showing more than 5 mm of offset shall be cause for rejection of that plate. New plates, if necessary, shall be field drilled using the hole pattern in the senior member as a template.

**421.48 Erection Tolerances –** Unless otherwise specified, the assembly tolerances and misalignment of members after erection shall be within the dimensional tolerances specified in CSA W59 and AASHTO LRFD Bridge Construction Specifications.

Misalignment shall be measured from vertical lines in the case of columns or towers, and from lines joining the ends of any test length of a member.

Joints, which are required on the drawings to be milled to bear, shall have at least 75% of the entire contact area in full bearing. The separations of any remaining portions shall not exceed 0.25 mm except locally at toes of flanges where a separation of 0.60 mm is permissible.

Where joints are not milled, the opening shall not exceed 13 mm.

**421.49 Field Painting –** This section applies to those parts of the steelwork which are to be painted. Coating Work shall be in accordance with SS 216 unless noted otherwise in SS 421. Build the coating using coatings from the specified SS 308 System. The coating system shall be from the same manufacturer as the shop coating system on the steel. After the completion of all deck and overhead concrete Work, steelwork shall be thoroughly cleaned of all rust, dirt, dust, oil and other foreign materials. Non-visible salts shall be removed in accordance with SSPC-SP WJ-4 to meet the NVC levels per SS 216.07.02(b). The shop coats of paint shall be touched up as necessary.

**421.49.01 Field Touchup –** Bare, rusty or damaged areas shall be cleaned to SSPC-SP11, Power Tool Cleaning to Bare Metal. Feather edges into the existing coating and build the coating as per SS 216.

After installation, Type 3 bolts, nuts and washers in the areas to be painted shall be cleaned to SSPC-SP11. These bolts, nuts and washers shall then receive the prime coat, stripe coat, midcoat and topcoat paint.

The masked off surfaces surrounding the bolt holes at connection locations shall be cleaned and shall receive the stripe coat, midcoat and topcoat paint.

**421.49.02 Topcoat –** The topcoat coat shall be chosen from either SS 308 System SS1 or SF2.

If the topcoat has not been applied in the shop, then the steelwork shall be given one topcoat coat of paint applied in accordance with the manufacturer's specifications and SS 216.

The colour of the finish coat will be selected by the Ministry. For weathering steel, unless noted otherwise, the topcoat colour shall match the expected colour of the oxidized surfaces (Federal Color 30045 unless noted otherwise). The proposed colour shall be subject to the acceptance of the Ministry Representative.

At the completion of the Contract, all steelwork, painted or unpainted, shall be cleaned of concrete spatter, mud, oil and other foreign materials.

**421.50 Unpainted Weathering Steel –** In the case of unpainted weathering steel, the outer faces of the girders and stringers, which includes the exposed edge of the top flange, the underside of the top flange, the girder web, and the top, bottom and outer edge of the bottom flange, and any other surfaces mentioned in the Drawings and Special Provisions, shall present a uniform surface free of mill scale and shall be sandblasted to SSPC-SP6 prior to installation. Cleaning shall also include all shop marks located on the exterior faces of the girders, and in all areas of interior girders that are readily visible to the public, as determined by the Ministry Representative. Concrete splatter adhering to the steel surfaces after the construction of the deck shall be removed and the steel surfaces cleaned.

**421.51 Touch-up of Galvanizing and Metallizing –** All field welds and other damage in galvanized and metallized coatings shall be touched up as follows:

If the Drawings or Special Provisions call for touch-up by metallizing, the damaged areas shall be locally sandblasted to "near white" per SSPC-SP10 / NACE No. 2. All dry abrasive blast cleaned areas shall be metallized in accordance with SSPC-CS23.00/AWS C2.23M/NACE No. 12, Specification for the Application of Thermal Spray Coatings (Metallizing) of Aluminum, Zinc, and Their Alloys and Composites for the Corrosion Protection of Steel, to provide a zinc coating not less than 0.3 mm in thickness.

Cut metal edges may be harder and consequently have less profile. If that is the case, they shall be ground back to softer metal and reblasted to achieve the specified profile.

Otherwise, the damaged areas shall be thoroughly cleaned and painted with two coats of Ministry-approved organic zinc-rich paint.

## QUALITY CONTROL PROGRAM

**421.61 Quality Control Program –** These Subsections describe the general Quality Control Program required by the Ministry for any Fabricator undertaking the fabrication of permanent steel bridges and steel bridge components. The

Quality Control Program shall be part of the overall Quality Management Plan for the project.

The term "Quality Control" defines those activities that the Fabricator performs to conform to the contract.

The term "Quality Assurance" defines those activities that the Ministry performs to audit the Contractor's quality control program and to ensure conformance to the Contract.

These Subsections contain statements of the Quality Objectives and Policies that the Ministry considers essential for successful and economical quality management. They also outline the Procedures and Documentation to implement and confirm that objectives are met.

The provisions of the Quality Control Program set forth in these Subsections shall apply to all steel bridges and bridge components contracted by or for the Ministry.

These Subsections require the establishment of a Quality Organization with the responsibility for the successful and timely implementation of all necessary Quality Control activities. Some positions shown in the Suggested Organization Chart may be held by the same individual. For example, the General Manager may also be the Contract Administrative Manager and the Purchasing Manager. The Plant Superintendent may also be the Receiver and the Welding Supervisor.

**421.62 Quality Objectives and Policies**

**421.62.01 Quality Objectives –** The Quality Objectives of the Ministry cover all steel bridges and steel bridge components produced under contract to the Ministry, as follows:

**(a)** Completed products shall conform fully to the governing Codes and Specifications stipulated in the Contract.

**(b)** The Quality Control Program shall be fully integrated into the ongoing manufacturing activities of the Fabricator.

**(c)** The operations of the Quality Control Program shall protect the interests of the Ministry with respect to scheduled delivery date and contracted price.

**421.62.02 Quality Policy –** The Quality Policy ensures that the product meets the Quality requirements of the Contract, is delivered on time, and is produced in a cost-effective manner.

The Quality Control Program applies to all stages of the design, drafting, procurement, manufacturing and testing of the product.

A Fabricator's Quality Control Manager shall be appointed with defined responsibilities in resolving quality matters and shall report to a senior management level. At each hold point, the Fabricator's Quality Control Manager shall:

**(a)** Document the successful completion of each stage as it progresses through fabrication and erection;

**(b)** Identify and report nonconforming components;

**(c)** Initiate or recommend disposition of nonconforming components;

**(d)** Verify corrections.

Any persons assigned to perform quality control inspections shall be other than those performing or directly supervising the Work and they shall not report directly to immediate supervisors responsible for producing the Work.

The Quality Control Program is not subordinate to any design, drafting, procurement, manufacturing or testing activities.

**421.63 Scope of the Quality Control Program –** The Quality Control Program governs the fabrication of steel bridges and bridge components for the Ministry. The Suggested Organization Chart is shown in Figure 421-1.

**421.63.01 Fabricator's General Manager Requirements –** The Fabricator's General Manager shall be responsible for:

**(a)** adhering to the Quality Control Program in all respects.

**(b)** ensuring that completed bridges or bridge components shall conform fully to the applicable design and to the fabrication and welding codes stipulated in the Contract.

**(c)** ensuring that all required documentation is produced according to the Quality Control Program.

**421.64 Range of Capability** – The Fabricator shall provide the necessary knowledge, skill (in-house or on a contract basis) and equipment to perform the following work on steel bridges and bridge components:

**(a)** Design of connections and joints not shown on Drawings (if applicable);

**(b)** Preparation of shop fabrication drawings;

**(c)** Preparation of bills of material;

**(d)** Preparation of material requisitions and purchase orders;

**(e)** Receiving, checking and storing materials for bridges;

**(f)** Layout, cutting, forming and fitting of parts;

**(g)** Assembly, tacking and welding;

**(h)** Dimensional checking and verification;

**(i)** Resolution of non-conformances;

**(j)** Documentation of all stages of work with capability of tracking all major components;

**(k)** Cleaning, painting, storing and shipping;

**(l)** Erection of bridges and bridge components (if applicable).

**421.65 Drawings and Specifications**

**421.65.01 Fabricator's Chief Draftsperson's Responsibilities –** The Fabricator's Chief Draftsperson shall:

**(a)** obtain the latest revision of the design drawings and specifications for the Work;

**(b)** submit shop drawings and erection drawings to the Ministry Representative for acceptance before commencement of the Work;

**(c)** prepare material requisitions containing a full description of the material sizes, material specifications, and certifications required for conformance to the Contract;

**(d)** deliver the material requisitions to the Purchasing Manager in ample time to permit ordering, delivery and documentation without delaying the Work;

**(e)** issue requisitions for all sub-contracted drafting work and shall ensure that all conditions of the Contract are part of such sub-contracts.

**421.65.02 Fabricator's Chief Design Engineer's Responsibilities –** The Fabricator's Chief Design Engineer shall be responsible for the design of any connections or joints not shown on the design Drawings; and for the design of the transportation details, erection procedures and any special erection equipment needed.

**421.66 Material Control –** All materials for the Work shall be ordered by the Fabricator's Purchasing Manager in full conformance with the material requisitions provided by the Fabricator's Chief Draftsperson.

The Purchase Orders shall contain all information necessary to ensure that materials purchased will comply fully with the terms of the Contract. Where mill certificates and test reports are required, it shall be so stated on the Purchase Order. Instructions shall state when the certificates and reports are to be delivered to the Fabricator.

If a supplier proposes a substitute for any material, the Fabricator's Purchasing Manager shall refer the proposed substitution to the Fabricator's Chief Design Engineer for review. If the substitute is acceptable to the Ministry Representative, the Fabricator's Chief Draftsperson shall amend all drawings and requisitions, withdraw old issues, and issue the new versions.

The Fabricator's Receiver shall:

**(a)** inspect all materials on arrival for conformance with the Purchase Orders;

**(b)** confirm that mill certificates and test reports are provided and that they correctly identify the materials delivered;

**(c)** arrange with the Fabricator's Plant Superintendent to store all materials for the Contract in segregated areas. Clear identification with the Contract shall be provided;

**(d)** issue a non-conformance report covering overage, shortage or damage to the materials, and copies of the report shall be provided to the Fabricator's Purchasing Manager and the Quality Control Manager.

The Fabricator's Purchasing Manager shall deliver all documentation to the Fabricator's Quality Control Manager for inclusion in the Quality Control file for the Contract.

**421.67 In-Progress Inspection and Reporting** – The Fabricator's Quality Control Manager shall ensure that only documented materials are used for the Contract.

All materials intended for incorporation into the Work shall be examined after cutting to size, forming and rolling. The Fabricator's Chief Inspector shall ensure conformance with the detailed shop drawings, shall report any non-conformance to the Fabricator's Quality Control Manager and shall order all work affected by the non-conformance to stop, pending approval of remedial action.

Before assembling any plate girders or complex parts, the surfaces of all materials shall be examined for imperfections revealed during previous fabrication operations. The joint edge preparation for all groove welds shall be verified as conforming to the Fabricator's CWB-approved welding standards and shall be within the acceptable tolerances.

After assembly of any plate girders or complex parts, and before commencing the strength welding, the assembly shall be checked for dimensional conformance. The Fabricator's Welding Supervisor shall ensure that the fit-up of all welded joints conforms to the approved welding standards.

The Fabricator's Quality Control Manager shall file a written report of verification with the Fabricator's General Manager and report any non-conformance.

**421.68 Correction of Non-conformance –** When a non-conformance is encountered, the Fabricator's Quality Control Manager shall determine a recommended disposition and obtain the Ministry's approval as quickly as possible.

If there is non-conformance to the material specification stipulated in the Purchase Order, the Fabricator's Purchasing Manager shall immediately find out the reasons for the delivery of non-conforming material. If the material is of a grade superior to that ordered, the Fabricator's Quality Control Manager and the Chief Engineer shall be notified. They shall verify that the material is an acceptable alternative in all respects, and this verification shall include consultation with the Ministry Representative. If the material is of a grade inferior to that ordered, it shall be rejected and the correct material or a superior material shall be obtained.

If there is non-conformance of material delivery that will delay production, the Contract Administration Manager and Fabricator's Plant Superintendent shall be notified immediately. They shall be given revised delivery dates for the adjustment of production scheduling. The Fabricator's General Manager shall determine the alternatives available and shall notify the Ministry Representative.

If there are non-conformities in material preparation, assembly, joint edge preparation and fit-up before strength welding, the Fabricator's Quality Control Manager and Welding Supervisor shall immediately review the non-conformance and notify the Fabricator's Chief Design Engineer who may require further investigation prior to submitting corrective action to the Ministry Representative for approval. If the necessary corrective action will result in delay to production, the Fabricator's General Manager shall be notified for adjustment to the production schedule. The Fabricator's Plant Superintendent shall inform the Fabricator's Quality Control Manager when the corrective actions are being done so that conformance can be verified and the non-conformance report cancelled.

**421.69 Welding –** All welding on structural and mechanical components shall be done by the company certified to CSA W47.1 (Division 1 or 2) and W186-M as applicable, and shall be done in accordance with the Fabricator's CWB-approved welding standards.

The edge preparations for all groove welds shall conform to the dimensions established in the approved welding standards.

If the joint is to be welded from one side only without back-gouging, the root gap and root face shall be checked to ensure conformance with the required geometry so that the root pass can be successfully completed.

The strength level and chemical composition of all filler materials used in structural and mechanical assemblies shall conform to the approved shop drawings.

All welding consumables shall conform to the approved welding standards and shall be received, stored and conditioned according to the applicable welding standards.

Any preheat required before welding shall be according to the approved welding standards.

The welding procedure followed in welding any joint in a structural or mechanical component shall conform to the applicable Data Sheet in the approved welding standards.

All welders and welding operators welding on structural or mechanical contracts shall be qualified under the requirements of the CSA Standard governing certification.

**421.70 Heat Treatment –** This covers any post-weld heat treatment necessary to conform to the approved welding standards.

**421.70.01 Fabricator's Chief Design Engineer's Responsibilities –** The Fabricator's Chief Design Engineer shall:

**(a)** decide whether any structural or mechanical components shall receive post-weld heat treatment to conform to the contract conditions or to the applicable Codes and Standards.

**(b)** inform the Fabricator's Chief Draftsperson of any such requirements so that they may be incorporated on to the approved shop drawings.

Any heat treatment stipulated on the approved shop drawings or contained in the approved welding standards shall be carried out at the appropriate time and according to the approved documents.

**421.70.02 Fabricator's Plant Superintendent's Responsibilities –** The Fabricator's Plant Superintendent shall ensure that any heat treatment stipulated is done according to the established procedures, shall obtain all documentation and reports, and shall deliver them to the Fabricator's Quality Control Manager.

**421.71 Non-destructive Testing –** The Ministry will test any weld by non-destructive testing methods, as described in SS 421.06.

The Fabricator's Plant Superintendent shall schedule the manufacturing operation to facilitate non-destructive testing.

If non-destructive testing of any welded joint reveals imperfections that are marginally more than the acceptance standards, the Fabricator's Quality Control Manager shall consult with the Ministry Representative regarding the location and nature of the imperfections. The effects of leaving minor defects in place shall be assessed in relationship to the loads carried by the joint and the possible adverse effects of making an unnecessary repair.

Weld repairs shall conform to the approved welding standards.

**421.72 Calibration of Measurement and Test Equipment –** All measurement and testing equipment owned by the Fabricator and used in the Quality Control Program shall be calibrated and re-calibrated at the intervals and in the manner stipulated in the Manufacturer's Instruction Manuals. Any adjusting devices shall be sealed or otherwise protected from unauthorized adjustment or tampering.

**421.73 Records Retention –** The Fabricator's record file for each Contract shall contain the pertinent drawings, purchase orders, bills of material, material mill certificates, test reports, Quality Control documents, NDT reports and certificates of compliance.

The Fabricator's record file shall be made available to the Ministry's Inspectors upon request.

Items in the Fabricator's record file shall be retained as per company policy.

The Fabricator's Quality Control Manager shall ensure that each file is complete in all respects before it is placed in the Company archives.

**421.74 Hold Points –** In planning the workflow, the Fabricator shall coordinate with the Ministry Inspector to decide "hold" points (as defined in SS 145.12) for inspection or non-destructive testing. A list of hold points shall be drawn

up by the Fabricator and the Ministry Representative at a prefabrication meeting. The Ministry Representative shall be informed of progress so that delays are minimized.

**421.74.01 Typical Hold Points –** The "hold" points will typically include some of, but not be limited to, the following:

**(a)** verification of materials

**(b)** after plate is prepared for splicing

**(c)** after splicing of plates

**(d)** after web to flange weld

**(e)** after stiffeners are applied

**(f)** camber of girders

**(g)** shop assembly

**(h)** cleaning and coating

**(i)** bearing plate attachment

**(j)** shipping arrangements

**(k)** subassemblies.

**421.74.02 Sign-Off –** Work shall not proceed past a "hold" point until it has been signed off by Quality Control and Quality Assurance. Reports shall be completed promptly.

**421.75 Transportation and Installation Procedures –** Transportation and installation procedures shall be prepared and submitted for review before any installation takes place on Site. The procedures shall be sealed by a professional engineer experienced in bridge erection and registered with the APEGBC.

Consideration shall be given to the following items during the preparation of the transportation and installation procedures.

**421.75.01 Girder Transportation**

**(a)** Brief description of hauling equipment;

**(b)** Location of girder support points;

**(c)** Engineering backup if supports vary from specifications; and

**(d)** Details of coating protection during loading transporting and erection.

**421.75.02 Installation Drawings**

**(a)** Bridge site plan showing piers, abutments and access roads;

**(b)** Crane make, crane chart, boom length(s) and crane locations;

**(c)** Mass of girder and access to crane(s); and

**(d)** Special installation equipment such as a launching truss, head frames and falsework.

**421.75.03 Commentary –** Brief point form description of installation sequence.

**421.75.04 Bearings –** Placement procedure for bearings to be included for multiple span continuous girders.

**421.75.05 Traffic Control –** Any arrangements that will be made for road and/or rail traffic.

**421.75.06 Utilities –** Safety and protection.

**421.75.07 Fall Protection –** Method and date of installation as required in specifications.

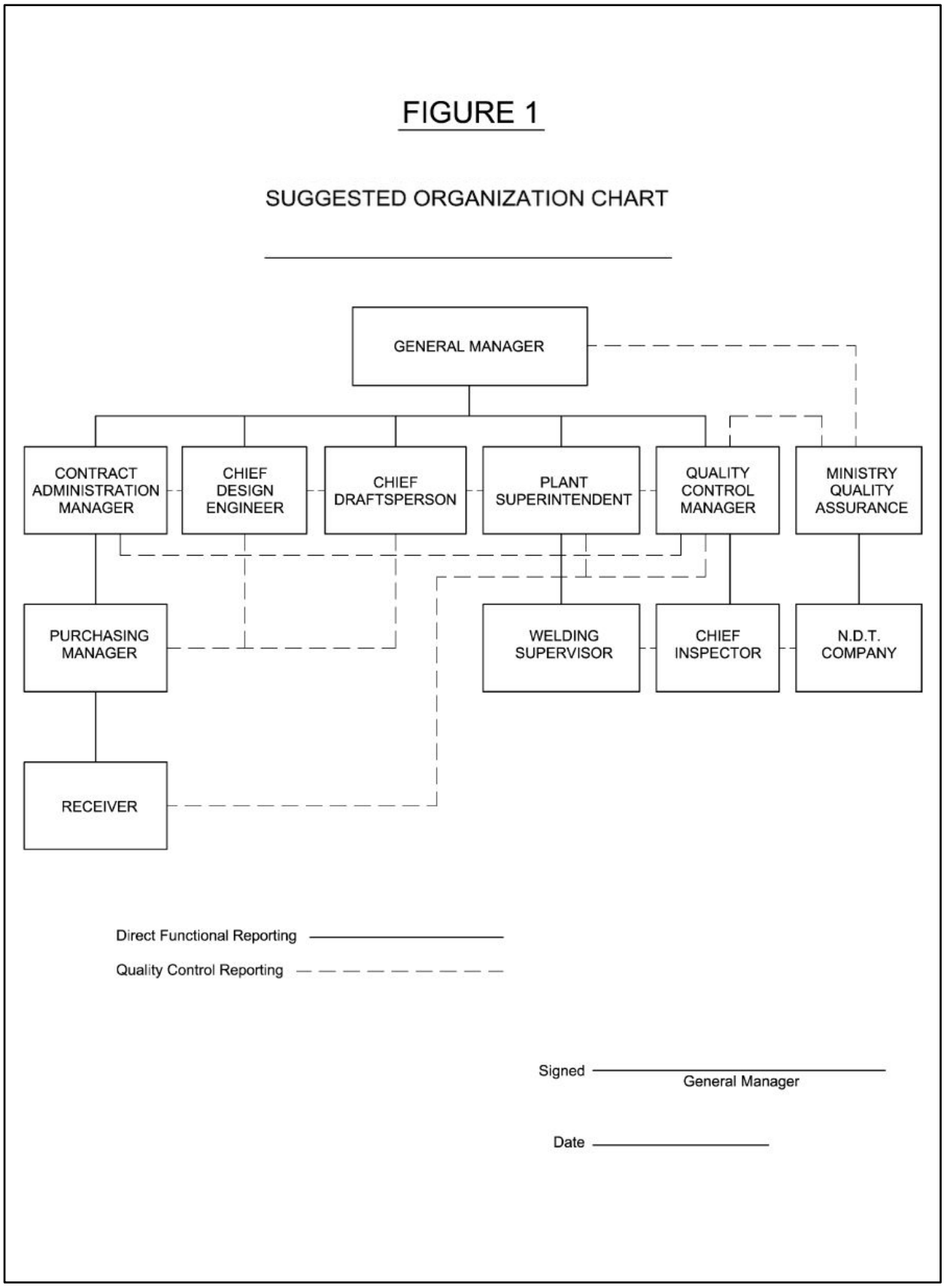
**PAYMENT**

**421.91 Supply and Fabrication –** Payment for supply and fabrication of structural steelwork will be made at the lump sum price bid. Payment shall be for quality control, working drawings, the supply, bending, cutting, assembly and welding of steelwork, including any galvanizing or metalizing required. Payment shall also cover trial assembly and storage as necessary.

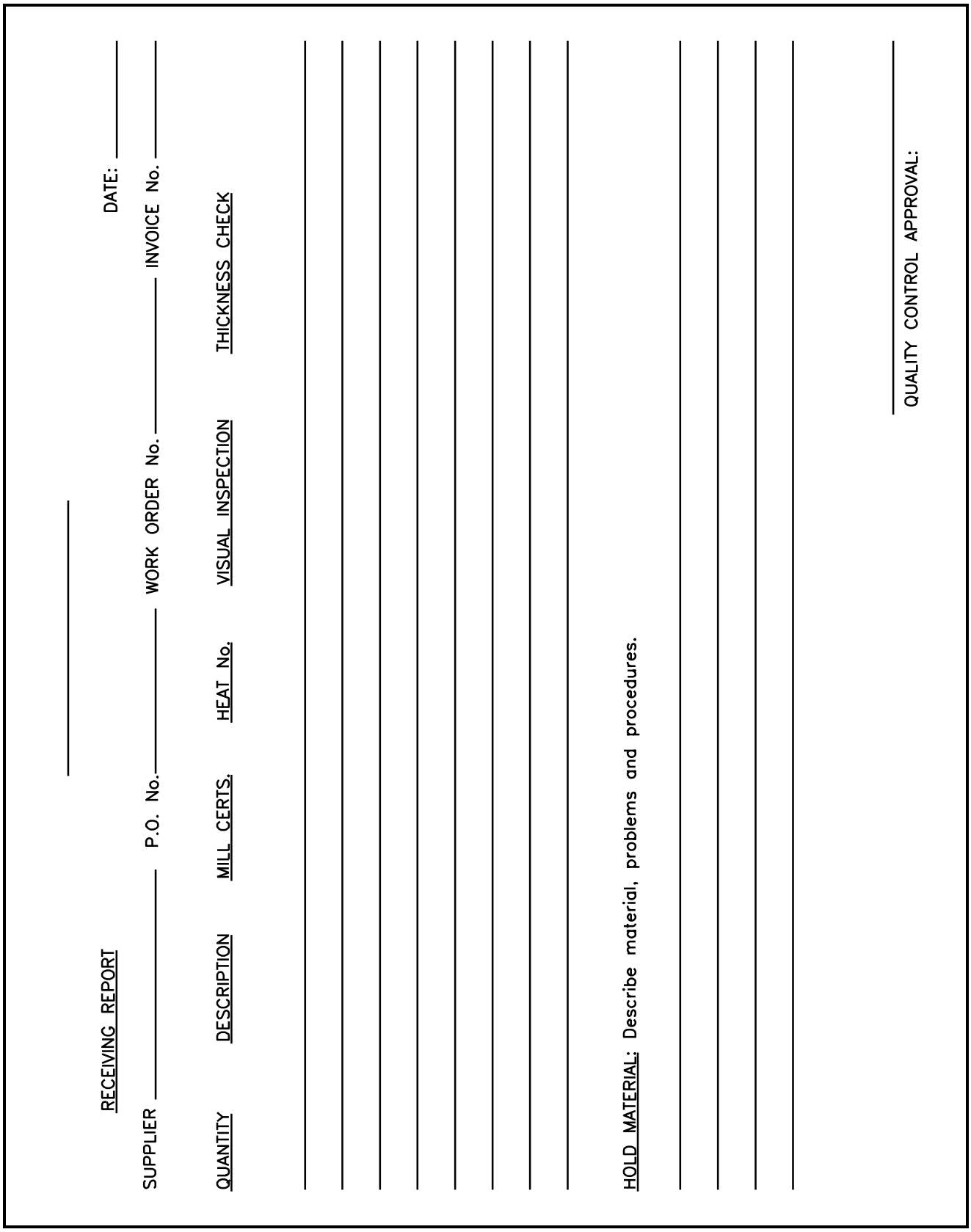
**421.92 Shipping and Erection** – Payment for shipping and erection of structural steelwork will be made at the lump sum price bid. Payment shall be for quality control and the loading, shipping and unloading of steelwork. Payment shall also cover falsework if necessary and the erection of steelwork.

**421.93 Painting –** Payment for painting of structural steelwork will be made at the lump sum price bid. Payment shall be for the surface preparation and the supply, application and touch-up of paint as required.

**Figure 421-1:Suggested Organization Chart**



**Sample Form 421-1: Receiving Report**



**Sample Form 421-2: Non-conformance Report**

## NON-CONFORMANCE REPORT

**BRIDGE NAME:** _____

**BRIDGE NUMBER:** _____

**JOB NUMBERS:** _____

**IDENTIFICATION DETAILS:** _____

_____

_____

**DESCRIPTION OF NON-CONFORMITY:**

_____

_____

_____

_____

_____, 20___ _____

Date Signature

**CORRECTIVE ACTION:**

_____

_____

_____

_____

**ACTION AGREED UPON**: _____ _____

**Quality Control Supervisor** **Quality Assurance Inspector**

**DATE COMPLETED:** _____, 20___ _____

Signature

**HOLD REMOVED BY:** _____ _____, 20___

Date

**QUALITY CONTROL SUPERVISOR:** _____ _____, 20___

Date

**QUALITY ASSURANCE SUPERVISOR:** _____ _____, 20___

Date

**Sample Form 421-3: Steel Fabrication Check List**

## Steel Fabrication Compliance Check List

**BRIDGE NAME:** _____

**BRIDGE NO.** _____

**MARK NUMBER(S):** _____

| | | Q.C. | Q.A. |
|---|---|---|---|
| Material: | Mill Certificates | | |
| | (Physicals, Chemicals, Impact Category) | | |
| | Identification on materials check | | |
| | Heat number transferred and recorded | | |
| Welding: | Procedures approved for each process | | |
| | Consumables checked | | |
| | Date Completed | | |

QC Inspector: _____ (Name) _____ (Signature) _____ (Date)

QA Inspector: _____ (Name) _____ (Signature) _____ (Date)

**Sample Form 421-4: Steel Fabrication Compliance Check List**

## Steel Fabrication Compliance Check List

BRIDGE NAME: _____ BRIDGE NO. _____

MARK NUMBER(S): _____

| | | Q.C. | Q.A. |
|---|---|---|---|
| Welding: | Weld preparation and fit up | _____ | _____ |
| | Welding within parameters | _____ | _____ |
| | Identification on materials check | _____ | _____ |
| | Shear studs | _____ | _____ |
| | Visual Inspection | _____ | _____ |
| | NDT outside agency results | _____ | _____ |

| | | Q.C. Req'd | QC Actual | Q.A. Verified |
|---|---|---|---|---|
| Dimensions: | Flanges Top Width 1 | _____ | _____ | _____ |
| | Width 2 | _____ | _____ | _____ |
| | Flatness | _____ | _____ | _____ |
| | Tilt | _____ | _____ | _____ |
| | Bottom Width 1 | _____ | _____ | _____ |
| | Width 2 | _____ | _____ | _____ |
| | Flatness | _____ | _____ | _____ |
| | Tilt | _____ | _____ | _____ |
| | Web Depth | _____ | _____ | _____ |
| | Section Length | _____ | _____ | _____ |
| | Bearing Ctr. Length | _____ | _____ | _____ |
| | Sole Plate Contract | _____ | _____ | _____ |
| | Camber | _____ | _____ | _____ |
| | Sweep Tolerance (1/1000) | _____ | _____ | _____ |
| | Splicing Camber | _____ | _____ | _____ |
| | Alignment | _____ | _____ | _____ |
| | Holes | _____ | _____ | _____ |
| | Date Completed | _____ | _____ | _____ |

**Sample Form 421-5: Coating Compliance Check List**

## Coating Compliance Check List

**BRIDGE NAME:** _____ **BRIDGE NO.** _____

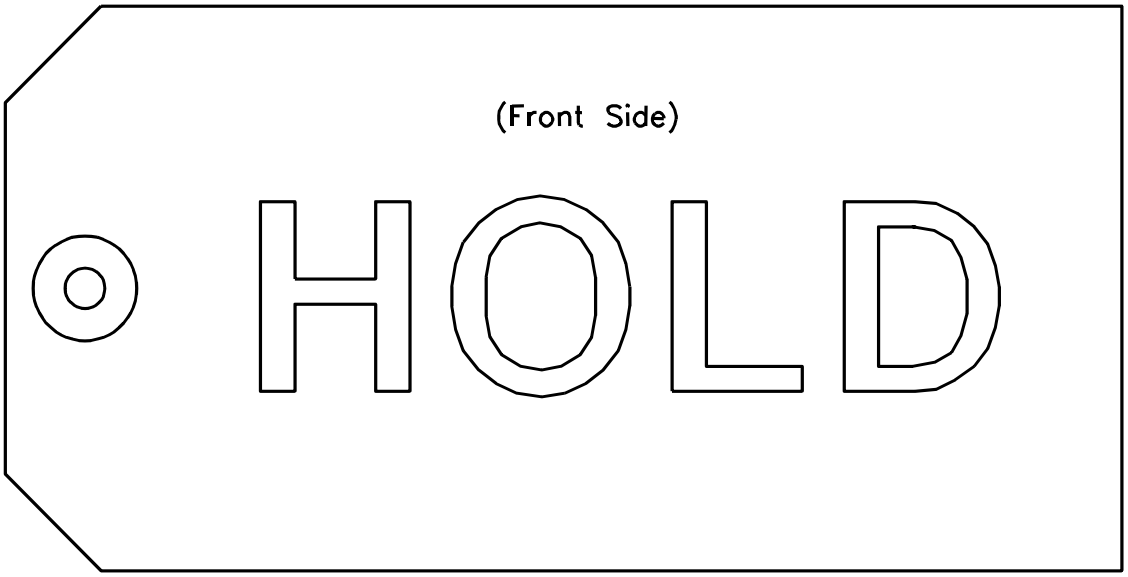
**MARK NUMBER(S):** _____

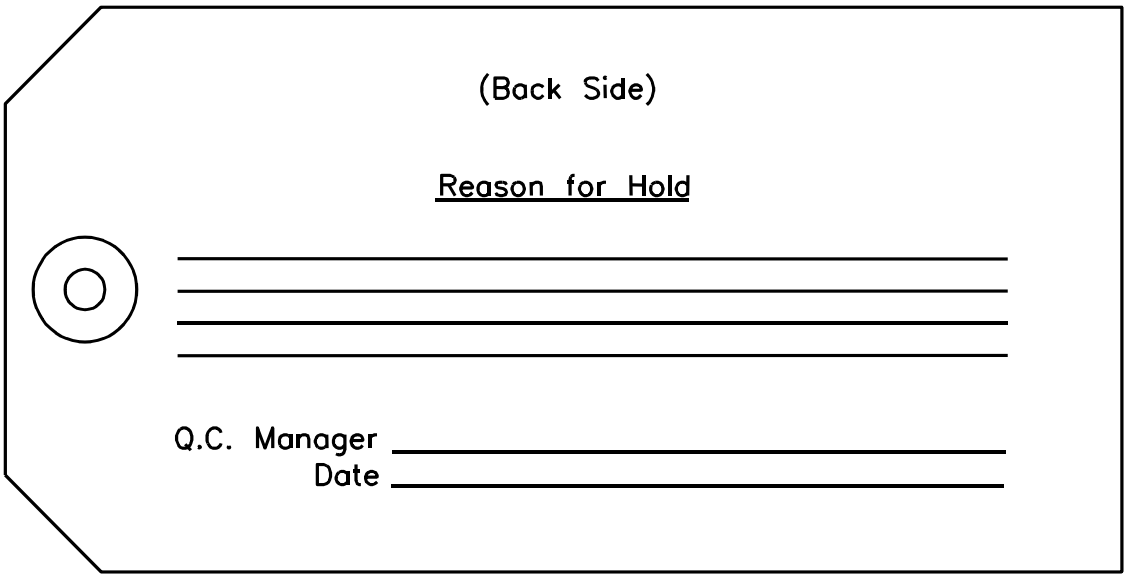
Check and sign below when inspected.

| | | Q.C. | Q.A. |
|---|---|---|---|
| Coating: | WHMIS Safety Data Sheets | _____ | _____ |
| | Technical Spec. Sheets | _____ | _____ |
| | Surface Preparation | _____ | _____ |
| | Ambient Conditions | | |
| | – Req'd | _____ | _____ |
| | – Actual | _____ | _____ |
| | Application | _____ | _____ |
| | Test Results | _____ | _____ |
| | Date Completed | _____ | _____ |

| | | | |
|---|---|---|---|
| **QC Inspector:** | _____ | _____ | _____ |
| | Name | Signature | Date |
| **QA Inspector:** | _____ | _____ | _____ |
| | Name | Signature | Date |

**Sample Form 421-6: Sample Tag**



# SECTION 422
# MISCELLANEOUS STEELWORK

**DESCRIPTION**

**422.01 Scope –** This Section covers steelwork apart from major structural steel elements (see SS 421).

**422.02 Adherence to Drawings –** Steelwork items shall be fabricated in strict accordance with the Drawings.

**422.03 Working Drawings –** The working drawings shall show all information necessary for the fabrication of the steelwork.

Working drawings shall be in the same system of units as the design Drawings.

Lettering for notes and dimensions shall be at least 2.5 mm and 4 mm for headings. Drawings shall be legible when printed on 11" x 17" sheets.

In the case of simple items, where the design Drawings are complete enough to be used for fabrication, the Contractor may, with approval of the Ministry Representative, dispense with working drawings. The Contractor shall notify the Ministry Representative of such intention in writing.

The Contractor shall submit to the Ministry Representative one complete set of all working drawings in digital format, for the Ministry's information only, two weeks prior to the fabrication of the steelwork. Prior to submission to the Ministry Representative, working drawings shall be reviewed and approved by the Contractor. By this review and approval, the Contractor represents that it has determined and verified all field measurements, field construction criteria, materials, and similar data, and that it has checked and coordinated each working drawing with the requirements of the Work and the contract documents. The Contractor shall indicate its review and approval by including on each drawing the date and signature of a person designated by the Contractor as being responsible for the Work.

The Contractor shall transmit working drawings through attachments to e-mail. Unless otherwise agreed to by the Ministry Representative, electronic attachments to an email must total no more than 20 MB and must be submitted unzipped. Drawing files shall be submitted in Portable Document Format (PDF) to print out on 11" x 17" size pages. PDF sets shall be created by "distilling" CAD sheets rather than by scanning paper plan sets. Unless otherwise agreed to by the Ministry Representative, electronic attachments greater than 20 MB in size shall be sent in two parts by separate emails, denoting "1 of 2" and "2 of 2" in the subject lines after other required subject-line information. Drawings shall be black images on a white background.

All PDF sheets shall be assembled within a single file, subject to the maximum file size indicated above, ensuring that all sheets are rotated to a "ready to read" orientation within the PDF file set. Generally, 11"x17" plan sheets should be in landscape and 8½" x 11" note sheets in portrait, so that the majority of text is vertical. PDF sheets shall be ready to print out on appropriately sized paper sheets with no additional formatting required by the viewer.

If agreed by the Ministry, the Contractor may submit shop drawings in paper format, in which case four complete sets of working drawings shall be submitted.

If agreed to by the Ministry Representative, the Contractor may employ a document and data management service such as SharePoint® to transmit working drawings. If this process is used, the limitation on drawing file size is waived. The Contractor shall be responsible for setting up the appropriate folders within the document and data management software and for providing access to these folders to the fabricator, Ministry Representative and the design engineer. Email notifications shall be sent to all parties whenever updates are made.

A copy of the shop drawings shall be available at all times at the location where the components shown on the drawings are being fabricated. Changes to the steelwork from what is shown on the shop drawings, or repairs made during fabrication and/or construction, shall be indicated by the Contractor on a marked-up set of shop drawings and submitted to the Designer and Ministry Representative at the completion of the Work.

**422.04 Quality Control –** The Contractor shall implement a quality control program to meet the Contract requirements. The quality control program shall be made available to the Ministry Representative for review.

**422.05 Quality Assurance** – The Ministry will implement a quality assurance program by auditing the quality control program and by inspection and testing at its discretion.

The Contractor shall notify the Ministry Representative at least 14 days before fabrication is to commence. In addition, the Contractor shall provide a minimum of 48 hours notice to the Ministry Representative that a product will be available for inspection. If the schedule is subsequently changed, the Contractor shall provide the 48 hours notice from the time that the Ministry is notified of this change. If the product is not available or is not sufficiently complete for inspection as notified, at the sole discretion of the Ministry Representative, the Contractor shall be charged stand-by costs for the Ministry's quality assurance inspector. The Contractor shall allow the Ministry Representative safe access to all parts of the Work and shall supply such information and assistance as is required. The Contractor shall provide samples of any materials, when requested by the Ministry. Inspection by

the Ministry shall not relieve the Contractor from obligation to perform the work in accordance with the Contract.

Any welding work found to be unacceptable shall be corrected in accordance with CSA W59, Section 5.10.

The Contractor shall provide 5 days notice to the Ministry Representative regarding intent to ship a unit or product and the product shall be made available for inspection prior to loading and shipping.

**422.06 Rejections –** The Ministry may reject any items which, in their opinion, do not comply with the requirements of the Contract. The Ministry in its sole discretion may back charge all inspection costs for the rejected material to the Contractor.

## MATERIALS

**422.11 Materials –** Structural steel, except where shown otherwise on the Drawings, shall conform to CSA-G.40.21, Grade 300 W or better.

The boron content of steel shall not exceed 0.0008%.

Chemical composition of steel to be galvanized shall be in accordance with SS 422.36 – Galvanizing.

The quality and care of electrodes shall conform to the requirements of the latest CSA W59.

**422.12 Storage of Materials –** Structural material shall be stored above ground. It shall be kept free from dirt and other foreign matter, and shall be protected as far as practicable from corrosion. Long members shall be supported on skids placed near enough together to prevent injury from deflection.

## CONSTRUCTION

**422.31 Welding –** All welding shall be done in accordance with procedures approved by the Canadian Welding Bureau (CWB) and reviewed by the Ministry.

All welded fabrication shall be done to the requirements of the latest CSA W59.

Welding shall be undertaken by a company approved by the CWB to the requirements of CSA W47.1, Division 2 or better.

**422.32 Welders and Welding Operators –** The Contractor shall produce evidence that all welders and welding operators to be employed on the work are currently qualified by CWB in the processes in which they are to be employed.

**422.33 Flame Cutting –** Steel may be oxygen or plasma cut, using a mechanical guide, provided a smooth surface is secured. Flame cutting by hand shall be done only where approved by the Ministry and the surface shall be made smooth by planing, chipping or grinding. The cutting flame shall be so adjusted and manipulated as to avoid cutting beyond the prescribed lines. Re-entrant flame cuts shall be filleted to a radius of not less than 20 mm. Edges of tension members shall be ground to a radius of 2 mm.

All exposed corners of members which are to be painted, shall be ground to a radius of 2 mm.

**422.34 Bent Plates –** When bending plates, the plates shall be so taken from the stock plates such that the bend line will be at right angles to the direction of rolling. Before bending, the corners of the plate shall be rounded to a radius of 2 mm throughout that portion of the plate at which bending is to occur. Bending shall be done by methods that will not crack, tear or otherwise injure the metal.

Cold bending and the minimum radii for cold bending shall be in accordance with CSA S6.

Hot bending of plates shall not be permitted without prior approval of the Ministry. Any proposed method for hot bending of plates shall be submitted to the Ministry for review and approval by the Ministry prior to proceeding with the work.

**422.35 Coating of Steelwork –** This subsection applies to those parts of the steelwork which are to be coated.

All coatings shall be applied according to their Manufacturer's product data sheet unless specified otherwise in the Contract Special Provisions.

Coating work shall be in accordance with SS 216 unless noted otherwise in this SS 422.

All exposed edges and corners of members, rolled or cut, which are to be coated, shall be ground to a minimum radius of 2 mm.

All steelwork which is to be coated, shall be given three shop coats of paint – primer, stripe coat and midcoat. Paint shall be chosen from SS 308 System SS1. Paint shall be supplied by the Contractor.

The topcoat may be applied in the shop or in the field. Field applied topcoat coat shall be chosen from either SS 308 System SS1 or SF2. If the topcoat is to be applied in the shop, then the Quality Control Program shall specifically address the integrity of the topcoat through to project completion.

Faying surfaces shall receive the prime coat only, as per SS 421.31.06. Faying surfaces shall not be coated with the stripe coat, midcoat or topcoat paint. Coating thicknesses and curing shall be those specified in the Manufacturer's Class B certificate.

Paint shall be applied in a covered area in accordance with the manufacturer's specifications and SS 216.

All edges, corners, crevices, bolts, nuts, protrusions and welds (unless ground flush) shall be stripe painted by brush for a width of 50 mm with the midcoat paint before the midcoat coat is applied. Stripe coating paint shall be applied as per SSPC-PA 1 and may be applied by spray, but shall be

brushed in. Stripe painting shall be allowed to cure before the midcoat coat is applied.

Surfaces shall be cleaned to "near-white" per SSPC-SP10 / NACE No. 2 and 50 $\mu$m to 75 $\mu$m (2 to 3 mils) sharp/angular profile.

The midcoat shall be applied to all surfaces including tops and sides of the top flanges, except faying surfaces. The midcoat shall not be applied until the primer and stripe coat are accepted by the Ministry Representative to be sufficiently cured.

Any surfaces inaccessible after erection, except faying surfaces, shall be given, in addition to the three shop coats, one coat of the topcoat paint appropriate for the paint system being used.

If the topcoat is to be applied in the shop, then the topcoat shall be applied after the midcoat is accepted by the Ministry Representative to be sufficiently cured.

The colour of the topcoat will be selected by the Ministry. The proposed colour shall be subject to the acceptance of the Ministry Representative.

**422.36 Galvanizing (if required) –** All steelwork to be galvanized shall be galvanized after complete fabrication to the requirements of ASTM A123 and ASTM A385.

**422.36.01 Embrittlement –** The galvanizer shall safeguard against embrittlement as required in ASTM A143. Galvanized members shall be subject, at the discretion of the Ministry, to the tests for embrittlement outlined in ASTM A143.

**422.36.02 Chemical Composition of Steel**

**(a)** The chemical composition of steel being galvanized shall be as follows:

- **(i)** Carbon less than 0.25%
- **(ii)** Phosphorus less than 0.04%
- **(iii)** Manganese less than 1.3%
- **(iv)** Silicon less than 0.04% or between 0.15% and 0.22%

**(b)** For steel not meeting these chemical composition requirements, special galvanizing techniques shall be developed by the galvanizer to ensure that the specified coating thickness and adherence is achieved. A detailed description of the special techniques shall be submitted to the Ministry Representative for review 2 weeks prior to galvanizing.

**422.37 Metallizing (if required) –** All surfaces to be metallized shall be treated after complete fabrication in accordance with SSPC-CS23.00 / AWS C2.23M / NACE No. 12, Specification for the Application of Thermal Spray Coatings (Metallizing) of Aluminum, Zinc, and Their Alloys and Composites for the Corrosion Protection of Steel.

The zinc coating shall not be less than 0.3 mm in thickness.

**422.38 Installation –** Steelwork shall be installed as shown on the Drawings.

Railings and fences shall be adjusted to produce uniform height and smooth alignment.

Field welding will be permitted only as shown on the Drawings.

**422.39 Field Painting (if required) –** This section applies to those parts of the steelwork which are to be painted.

Coating work shall be in accordance with SS 216 unless noted otherwise in this SS 422.

All steelwork shall be thoroughly cleaned of all rust, dirt, dust, oil and other foreign materials. Non-visible salts shall be removed in accordance with SSPC-SP WJ-4 / NACE WJ-4 to meet non-visible contaminant (NVC) levels per SS 216.07.02(b). The shop coats of paint shall be touched up as necessary.

**422.39.01 Field Touch-up –** Bare, rusty or damaged areas shall be cleaned to SSPC-SP 11, Power Tool Cleaning to Bare Metal. Feather edges into the existing coating and build the coating as per SS 216. Build the coating using coatings from the specified SS 308 System. Coating system shall be from the same manufacturer as the shop coating system on the steel.

**422.39.02 Topcoat –** The topcoat coat shall be chosen from either SS 308 System SS1 or SF2. If the topcoat has not been applied in the shop, then the steelwork shall be given one topcoat coat of paint applied in accordance with the manufacturer's specifications and SS 216.

The colour of the finish coat will be selected by the Ministry. The proposed colour shall be subject to the acceptance of the Ministry Representative

**422.40 Touch-Up of Galvanizing and Metallizing –** All field welds and other damage in galvanized and metallized coatings shall be touched up as follows:

If the contract documents call for touch-up by metallizing, the damaged areas shall be locally cleaned to the requirements of SSPC-SP 5 or SSPC SP 11, with an angular profile of 50 to 100 µm. All cleaned areas shall be metallized in accordance with SSPC-CS23.00 / AWS C2.23M / NACE No. 12, Specification for the Application of Thermal Spray Coatings (Metallizing) of Aluminum, Zinc, and Their Alloys and Composites for the Corrosion Protection of Steel, to provide a zinc coating not less than 0.3 mm in thickness.

Cut metal edges may be harder and consequently have less profile. If that is the case, they shall be ground back to softer metal and reblasted to achieve the specified profile.

Otherwise, the damaged areas shall be thoroughly cleaned, then painted with two coats of Ministry-approved organic zinc-rich paint.

**422.41 Clean-up –** At the completion of the Contract, all miscellaneous steelwork shall be cleaned of concrete spatter, mud, oil, shop markings if visible to the public and other foreign materials.

**PAYMENT**

**422.91 Payment –** Payment for miscellaneous steelwork will be made at the Lump Sum price(s) bid. Payment shall be for shop drawings and the supply, fabrication, galvanizing and/or painting if required and installation of the steelwork.

# SECTION 502

# ASPHALT PAVEMENT CONSTRUCTION
# END PRODUCT SPECIFICATION (EPS)

## GENERAL

**502.01 Preliminary and General –** This Section describes the materials, equipment, professional standards, and end product requirements for the construction of hot mix asphalt pavement and related shouldering operations.

**502.01.01 End Product Specifications (EPS)** – Specifications containing the acceptance and payment criteria based on the results of specified sampling and testing. Payment of the Contract Unit Prices for the asphalt pavement product is contingent on the product meeting the Quality Control (QC) Plan, professional standards and quality requirements of this Section and is subject to payment adjustments upward and downward in accordance to the provisions provided in this Section.

**502.01.02 Acceptance and Authorization –** When used in this Section and subject to the General Conditions:

**(a)** **"Acceptance"** means agreement with past actions or decisions made, within the scope of the Contract.

**(b)** **"Authorization"** means formal approval for future actions, frequently changing the Contract requirements.

**502.02 General Description of the Work –** Generally, the work associated with the construction of asphalt pavement and shouldering by EPS consists of the following:

**(a)** Preparing a QC Plan for review before commencing the Work and providing at the production site a testing facility to provide the data needed to implement that plan;

**(b)** Supplying, screening, crushing, processing and improving aggregate to produce asphalt mix aggregate;

**(c)** Supplying and delivering asphalt cement and spray primer meeting the requirements of SS 952;

**(d)** Preparing mix designs which, once reviewed and accepted by the Ministry Representative, become the basis for the accepted Job Mix Formula;

**(e)** Heating the asphalt mix aggregate and mixing it with asphalt cement to produce asphalt mix that meets the Job Mix Formula;

**(f)** Hauling, placing, compacting and finishing the asphalt mix; and

**(g)** Supplying, hauling, placing and compacting shoulder gravels.

The Contractor shall provide a paving end product conforming in professional standard, quality and accuracy of detail to the QC Plan and the dimensional and tolerance requirements of the contract. Where no tolerances are specified, the standard of workmanship shall be in accordance with normally accepted good practice and the provisions of this Section. Payment is subject to upward or downward adjustments based on quality acceptance tests performed by the Ministry Representative and calculations performed by the Contractor with respect to application rate.

**502.03 Definitions**

**(a)** **Actual Asphalt Content –** The amount of asphalt cement in the mix as determined by the Ministry's Quality Assurance Program.

**(b)** **Additives –** Solid or liquid materials to enhance the properties of the liquid asphalt cement or mix.

**(c)** **Aggregate –** The crushed or screened gravel.

**(d)** **Asphalt Cement (AC) –** A bitumen-based liquid binder used in asphalt pavement.

**(e)** **Asphalt Content** – The percentage of asphalt cement in the mix expressed as percentage by weight of the total dry aggregate in the mix determined by the Oven Test procedure.

**(f)** **Asphalt Mix (AM)** – Hot plant mixture of asphalt cement and aggregate.

**(g)** **Asphalt Mix Aggregate (AMA) –** The processed crushed aggregate just prior to the addition of asphalt cement.

**(h)** **Asphalt Mix Design (AMD)** – The asphalt mix design that is developed through the initial trials and testing to determine and optimize the Job Mix Formula for the end product of the asphalt mix.

**(i)** **Asphalt Pavement (AP) –** Compacted asphalt mix.

**(j)** **Cutback Asphalt –** Asphalt cement which has been blended with light petroleum distillates.

**(k)** **Design Asphalt Content –** The asphalt content upon which the Job Mix Formula is initially established.

**(l)** **Driving Lane –** A single lane in any area of the pavement other than a shoulder or a barrier flare.

**(m)** **Emulsified Asphalt –** Asphalt cement that has been blended with water and emulsifying agents to form aqueous emulsions, including anionic type, cationic type and high float type.

**(n)** **End Product Specification (EPS) –** A specification whereby the Contractor is responsible for the workmanship and quality control of the construction processes, and whereby the Ministry reviews the

workmanship and may perform the specified quality assurance sampling and testing of the end product for the purpose of determining acceptance/rejection and payment.

**(o) Job Mix Formula (JMF) –** The asphalt mix "recipe", proposed by the Contractor in accordance with SS 502.08.04 or an accepted variation in accordance with SS 502.08.10 and accepted by the Ministry, establishing the aggregate proportions, gradation, and the asphalt content to be used for production of asphalt mix.

**(p) Levelling Course (LC) –** Asphalt mix used to improve crossfall, level, and strengthen existing pavements.

**(q) Lift** – A layer of asphalt mix laid in a single application then compacted.

**(i) Top Lift** The uppermost Lift, forming the final running surface.

**(ii) Lower Lift** Any Lift below Top Lift.

**(iii) Bottom Lift** The lowest Lift (excluding Level Course),

**(r) Lot and Sub-Lot –** A Lot is a portion of the work being considered for acceptance and for the determination of payment adjustments. For density, AC content, gradation and smoothness, each Lot is comprised of a number of Sub-Lots, each of which is sampled, and then aggregated to determine the acceptability of the Lot. Lot and Sub-Lot sizes are defined in the appropriate payment adjustment provisions of this Section.

**(s) Reclaimed Asphalt Pavement (RAP) –** Asphalt Pavement that has been removed and processed, for the purpose of recycling.

**(t) 502.03.20 Recycled Asphalt Mix (RAM) –** A carefully controlled hot plant mix of asphalt cement, graded high quality aggregate, and reclaimed asphalt pavement.

**(u) Reject Mix –** Asphalt mix that is deemed unacceptable for use in the project.

**(v) Sample Mean –** The arithmetic mean of a set of test results constituting the sample.

**(w) Smoothness –** A measure of the longitudinal profile of the pavement surface, measured as International Roughness Index (IRI).

**(x) Surplus Aggregate –** Aggregate surplus to the works, in split or un-split stockpiles which singly or combined will meet the asphalt mix aggregate gradation.

**(y) Random Sample –** A set of test measurements taken, one each from a number of separate areas or Sub-Lots within a Lot, in an unbiased way.

**(z) Voids in Mineral Aggregate (VMA)** – The space available to accommodate the effective volume of asphalt (asphalt not absorbed into the aggregate) and the volume of air voids necessary in the mixture.

**502.04 Quality Control**

**502.04.01 General –** Quality Control shall be performed by the Contractor in accordance with SS 101 and the requirements of this Section.

**502.04.02 Quality Control (QC) Plan –** The paving component of the Contractor's Quality Control Plan shall fulfill or exceed the requirements of APPENDIX 502-A and function integrally with any other Quality Management provisions of the Contract.

The QC Plan shall be submitted in accordance with the timelines established in SS 101.02.03(c).

The QC Plan must include a detailed description of the means by which the Contractor shall use the quality control test results to ensure that the workmanship, asphalt materials, aggregate, mix production, paving and pavement compaction processes will be controlled to keep the product within the specified limits. The QC Plan must clearly show the flow of information from the quality control laboratory to the individuals who shall make the actual adjustments to the processes and equipment to affect this control. The plan will show time allowance for each step, the names and positions of all the people involved, and a clear description of the responsibilities of each.

**502.04.03 Quality Control Testing and Inspection –** The Contractor shall provide and maintain equipment and qualified personnel to perform all laboratory testing, field testing and inspection necessary to determine and monitor the characteristics and properties of all the materials produced and incorporated into the work. They shall also monitor the workmanship of the final product in accordance with the QC Plan as most recently submitted and accepted.

The Contractor shall provide a testing facility(s) that meets the requirements necessary to carry out all the test procedures listed within this Section. The facility(s) must have the equipment specified under the appropriate test designation to perform the tests. The Ministry shall have access at all times to the quality control facility(s).

**502.04.04 Quality Control Records** – Quality Control inspections shall be recorded on check sheets and/or diaries at the time of inspection.

The results from Quality Control testing shall be reported on test logs and plotted on charts immediately after each test is completed. The Contractor shall report all test results on Ministry-supplied forms (available from the Ministry Representative) or Contractor-supplied forms acceptable to the Ministry Representative.

For the purposes of confirming delivery of asphalt mix and shouldering aggregate to the road and the calculation of

material application rates, the Contractor shall provide to the Ministry Representative a copy of the Road Checker's Summary and the weigh tickets for each load received at the placement operation at the end of each shift. The Road Checker's Summary shall include, but not be limited to, the following information:

**(a)** Truck Number;

**(b)** Weigh Ticket Number and Net Weight of load;

**(c)** Date, time and location by station of delivery;

**(d)** Material Application Rate Dimensions and Calculations. Calculation frequency minimally shall be for every 10 loads; and

**(e)** Notes pertaining to the paving of any appurtenances (letdowns, intersections, tapers, etc.).

In addition to the equipment calibration requirements of SS 101, for the purpose of obtaining accurate and consistent results between the Contractor's Quality Control testing and the Ministry's Quality Acceptance testing, correlation of the Marshall hammer and ignition ovens to be used on the project shall be performed prior to mix production. Correlation procedures to be followed for both the Marshall hammer and ignition oven are included as Appendices to this SS 502.

**502.04.05 Final Quality Control Testing Reports –** Prior to the issuance of a Completion Certificate, the Contractor shall provide the Ministry Representative with:

**(a)** A summary of all aggregate quality control test results;

**(b)** Copies of all quality control test results for asphalt mix properties and compaction; and

**(c)** Copies of all quality control charts.

## MATERIALS

### 502.05 Materials Testing

**502.05.01 Test Procedures –** Where a test is specified to conform to an ASTM procedure, the listed corresponding AASHTO test may be used, or vice versa. See Table 502-A.

**502.05.02 Sieve Sizes –** All aggregate gradation tests shall use the following sieve sizes: 37.5 mm, 25 mm, 19 mm, 16 mm, 12.5 mm, 9.5 mm, 4.75 mm, 2.36 mm, 1.18 mm, 0.600 mm, 0.300 mm, 0.150 mm, and 0.075 mm.

### 502.06 Aggregates

**502.06.01 Aggregates –** The Contractor shall supply all aggregates for the Work from sources acceptable to the Ministry Representative.

**502.06.02 Work in Ministry Pits or Quarries –** Ministry pits offered in the Special Provisions as available sources may be used without charge provided they are developed and worked in accordance with SS 145.26, SS 202 Part C, and the pit development plan. Deviation from the pit development plan will require the authorization of the Ministry Representative. No guarantee is given or implied that aggregate from a Ministry pit(s) will meet the specification requirements or provide requisite quantities.

Without limitation and unless otherwise specified in the Special Provisions, the following items shall not be left in a Ministry pit:

**(a)** Surplus milled or rubblized asphalt pavements;

**(b)** Reject mix;

**(c)** Fuel contaminated materials; or

**(d)** Other waste products.

**502.06.03 Supply of Aggregates, Aggregate Production and Characteristics –** The Contractor shall not produce aggregate until the Contractor has received written notification from the Ministry Representative that their QC Plan is in accordance with SS 502.04.02, and has in place testing facilities for aggregate production that are in accordance with the QC Plan.

For the production of aggregate within Ministry pits, the Contractor shall follow the Ministry's pit development plan, and provide crushing equipment such that all aggregate, which will pass through 375 mm x 450 mm slotted openings, shall be used for the production of crushed aggregate. Rocks, which will not pass through these openings, shall be stockpiled safely at a location in the pit as directed by the Ministry Representative. Crushing and screening equipment shall be provided with adequate facilities and capacity to be able to bleed off reject aggregate and remove any excess fine aggregate, dust or objectionable aggregate coatings, to make it generally acceptable for use. No portion of the products from crushing or screening plants that can be used shall be wasted but shall be stockpiled or used as directed by the Ministry Representative.

Where the Ministry has any available test result information on the properties shown in Table 502-B, for a Ministry pit or other source, the Ministry Representative will upon request, provide that information to the Contractor. Otherwise, sampling and testing to determine and demonstrate the compliance of aggregate with the requirements of this Section shall be the responsibility of the Contractor.

**502.06.04 Shouldering Aggregate –** Shouldering aggregate shall meet the following unless otherwise specified in the Special Provisions or authorized by the Ministry Representative:

**(a)** Aggregate may be AMA, 25 mm WGB, or HFSA; and

**Table 502-A: Standard Sampling and Testing Procedures**

| Test Ref. No. | | Standard Test Method for… (ASTM test title, unless otherwise noted) |
|---|---|---|
| **AASHTO** | **ASTM** | |
| **Aggregates:** | | |
| T 11 | C117 | Materials Finer than 75 μm (No. 200) Sieve in Mineral Aggregates by Washing |
| T 85 | C127 | Relative Density (Specific Gravity), and Absorption of Coarse Aggregate |
| T 84 | C128 | Relative Density (Specific Gravity), and Absorption of Fine Aggregate |
| T 27 | C136 | Sieve Analysis of Fine and Coarse Aggregates |
| T 112 | C142 | Clay Lumps and Friable Particles in Aggregates |
| T 255 | C566 | Total Evaporable Moisture Content of Aggregate by Drying |
| R 90 | D75 | Standard Practice for Sampling Aggregates |
| T 304 | C1252 | Standard Test Methods for Uncompacted Void Content of Fine Aggregate (as Influenced by Particle Shape, Surface Texture, and Grading) [Method "A"] |
| | D421† | Standard Practice for Dry Preparation of Soil Samples for Particle-Size Analysis and Determination of Soil Constants [aka Atterberg Limit] |
| T 176 | D2419 | Sand Equivalent Value of Soils and Fine Aggregate |
| | D4318 | Standard Test Methods for Liquid Limit, Plastic Limit, and Plasticity Index of Soils |
| | D5821 | Determining the Percentages of Fractured Particles in Coarse Aggregate |
| T 327 | D6928 | Resistance of Coarse Aggregate to Degradation by Abrasion in the Micro-Deval Apparatus |
| **Asphalt Cement:** | | |
| T 49 | D5 | Penetration of Bituminous Materials |
| T 202 | D2171 | Viscosity of Asphalts by Vacuum Capillary Viscometer |
| T 350 | D7405 | T 350 Standard Method of Test for Multiple Stress Creep Recovery (MSCR) Test of Asphalt Binder Using a Dynamic Shear Rheometer (DSR) |
| **Asphalt Mix and Pavement:** | | |
| | D979 | Standard Practice for Sampling Bituminous Paving Mixtures |
| T 283 | D1075 | Effect of Water on Compressive Strength of Compacted Bituminous Mixtures |
| | D1188 | Bulk Specific Gravity and Density of Compacted Bituminous Mixtures Using Paraffin Coated Specimens |
| | D2041 | Theoretical Maximum Specific Gravity and Density of Bituminous Paving Mixtures |
| | D2726 | Bulk Specific Gravity and Density of Non-Absorptive Compacted Bituminous Mixtures |
| T 269 | D3203 | Percent Air Voids in Compacted Dense and Open Bituminous Paving Mixtures |
| | D4469 | Calculating Percent Asphalt Absorption by the Aggregate in an Asphalt Pavement Mixture |
| T 245 | D5581 | AASHTO Standard Method of Test for Resistance to Plastic Flow of Bituminous Mixtures Using Marshall Apparatus |
| T-308 | D6307 | Asphalt Content of Hot Mix Asphalt by Ignition Method |
| | D6927 | Marshall Stability and Flow of Asphalt Mixtures |
| **Shouldering:** | | |
| | D698 | Standard Test Methods for Laboratory Compaction Characteristics of Soil Using Standard Effort (12,400 ft-lbf/ft$^3$ (600 kN-m/m$^3$)) [aka Standard Proctor] |
| | D5220 | Standard Test Methods for Water Content of Soil and Rock in Place by Neutron Depth Probe Method |
| | D6938 | In-Place Density and Water Content of Soil and Soil- Aggregate by Nuclear Methods (shallow depth) |

*Note: † – Practice has been "withdrawn" by ASTM, but last published version D421-85(2007) is still applicable to the Work.*

**Table 502-B: Requirements for Coarse Aggregate**

| Test Ref. # | | Procedures | Superpave | Class 1 | Class 2 |
|---|---|---|---|---|---|
| AASHTO | ASTM | | | | |
| T 85 | C127 | Maximum Water Absorption, % by mass | 2 | 2 | 2 |
| T 112 | C142 | Maximum % by mass of clay balls and friable particles | 1.0 | 1.0 | 1.5 |
| | D5821 | 2 Fractured Faces, Minimum % by Mass retained on the 4.75 mm sieve | 90 | 85 | 70 |
| T 327 | D6928 | Maximum Micro-Deval abrasion loss factor, % | 18 | 18 | 20 |

**(b)** Aggregate gradations and physical properties shall comply with:

**(i)** SS 502.06, if AMA is to be used; or

**(ii)** SS 202.04 and SS 202.05, for any other aggregates.

**502.06.05 Paving Aggregates –** Paving aggregates shall meet the following requirements:

**(a) Coarse Aggregates**

**(i)** Shall be all mineral matter retained on the sieve designated in the test procedures for each individual test.

**(ii)** Shall consist of crushed stone, crushed gravel, or combination thereof, or materials naturally occurring in a fractured condition, or materials naturally occurring of highly angular nature or rough texture.

**(iii)** Shall be free from coating of clay, silt or other deleterious material.

**(iv)** Shall meet the requirements listed in Table 502-B.

**(b) Fine Aggregate**

**(i)** Shall be all mineral matter passing the sieve designated in the test procedure(s) for each individual test.

**(ii)** Shall be clean, tough, durable, moderately sharp, and free from coatings of clay, silt, or other deleterious material, and shall contain no clay balls or other aggregations of fine material.

**(iii)** Shall have a sand equivalent of not less than 40 (not less than 45 for Superpave mixes) when tested in accordance with ASTM D2419.

**(iv)** For Class 1 and Superpave mixes, shall have a minimum value of 45 when tested according to the AASHTO T 304, Method "A", - Uncompacted Void Content of Fine Aggregate when determining Fine Aggregate Angularity.

**(c) Mineral Filler and Mineral Dust**

**(i)** Mineral filler shall consist of all matter passing the 0.600 mm sieve and mineral dust shall consist of all mineral matter passing the 0.075 mm sieve.

**(ii)** Shall both be free from organic matter.

**(iii)** Mineral filler shall be non-plastic when tested in accordance with ASTM D4318.

**(d) Superpave Aggregates** – Aggregates for Superpave mixes shall have properties and the gradation limits as specified below and in accordance with the latest version of the Asphalt Institute's *MS-2 Asphalt Mix Design Methods (current version)*. Changes and/or variations from these limits shall be outlined within the Special Provisions.

**(i)** 90% fractured aggregate with a 12.5 mm nominal maximum size, including sufficient manufactured fines to provide fine aggregate angularity.

**(ii)** The aggregates must meet all the requirements for angularity, toughness deleterious materials, clay content, and flat and elongated particles.

**(iii)** Design ESALs will be 10 – 30 million.

**502.07 Supply of Asphalt Cement and Primer –** The Contractor shall supply the types and grades of asphalt cement and primers as specified in the Special Provisions and in accordance with SS 952. The supply of these materials includes, but is not limited to, ordering, scheduling delivery of, receiving, handling, storing, sampling, and testing of the materials and other related work.

The Contractor shall supply the Ministry Representative with copies of the asphalt supplier's weigh-bill and records of all asphalt materials received on a daily basis.

**502.07.01 Asphalt Binder Testing** – The Contractor shall provide supplier's Asphalt Binder testing and grade information upon request.

**502.07.02 Testing for Viscoelastic Properties** – The Ministry may test an Asphalt cement to determine its viscoelastic properties.

The selected grades of Asphalt Cement may be tested at a temperature of 58°C to determine the average percent recovery at 3.2 kPa ($R_{3.2}$) according to the requirements of ASTM D7405.

The minimum $R_{3.2}$@58°C value for selected asphalt binder grades shall be determined as outlined in Table 502-C.

**Table 502-C: MSCR – Elastic Recovery Requirements**

| PGAC | Minimum $R_{3.2}$ @ 58°C |
|---|---|
| 58-34<br>64-28 | 25% |
| 58-37<br>58-40<br>64-34<br>70-28 | 40% |
| 64-37<br>76-28 | 55% |

**502.08 Asphalt Mix**

**502.08.01 Responsibility for Asphalt Mix Design** – Preparation and submission of the project asphalt mix designs for Ministry review is the responsibility of the Contractor. All costs incurred in mix design formulation are the responsibility of the Contractor.

The Contractor shall utilise a qualified registered member of the Association of Professional Engineers and Geoscientists of British Columbia or a qualified, registered member of the Applied Science Technologists and Technicians of British Columbia who shall sign off the asphalt mix design. The Contractor shall utilise a CCIL certified testing laboratory for

**(a)** meeting the requirements of SS 101;

**(b)** assessing the aggregate material proposed for use; and

**(c)** preparing Asphalt Mix Designs as required in SS 502.

**502.08.02 Requirements for Asphalt Mix Design**

**(a)** **All Mixes –** Asphalt mix designs shall be performed using the asphalt cement grade specified in the Special Provisions and which is from the same refinery contracted to supply the asphalt cement for the duration of the project.

Any subsequent change in the asphalt cement supplied by the Contractor will require a new asphalt mix design unless permitted otherwise by the Ministry Representative.

**(b)** Asphalt mix designs, Job Mix Formulas, and field adjustments made in accordance with SS 502.08.10 must all be based on an aggregate gradation meeting the requirements of Table 502-D and the criteria specified in Table 502-E or Table 505–F, as applicable.

**(c)** Marshall Mixes – The asphalt mix design for Class 1 and Class 2 pavements shall be carried out under Marshall design criteria using the designated equipment and procedures as contained in the Asphalt Institute's "*MS-2 Asphalt Mix Design Methods*" (current version). Marshall hammers shall be correlated in accordance with APPENDIX 502-F.

**(d)** **Superpave Mixes –** The Superpave asphalt mix design shall be carried out in accordance with the latest edition of the Asphalt Institute's "*MS-2 Asphalt Mix Design Methods*" (current version) and this specification.

The JMF aggregate gradation shall fall under the maximum density gradation line for the 0.300 mm to 2.36 mm sieves, inclusive.

**(e)** **Open Graded Friction Course ("OGFC") Mixes –** The OGFC asphalt mix design shall be carried out under the Marshall design criteria using the designated equipment and procedures in accordance with the latest edition of the Asphalt Institute's "*MS-2 Asphalt Mix Design Methods*" (current version).

**502.08.03 Asphalt Mix Anti-strip Additives**

**(a)** **All Mixes –** An anti-stripping agent, compatible with the supplied asphalt cement, shall be chosen by the Contractor from the Ministry's Recognized Products List, and shall be added to all asphalt mix used in the Work. The Recognized Products List can be found online at:

https://www2.gov.bc.ca/gov/content/transportation/transportation-infrastructure/engineering-standards-guidelines/recognized-products-list

The Contractor shall notify the Ministry Representative of the type of anti-stripping agent and the percent by weight being added.

For all mix designs, the Contractor shall determine the Tensile Strength Ratio (TSR) of each asphalt mix in accordance with AASHTO T 283.

Asphalt cement must still meet the specified grade following the addition of the anti-stripping agent.

**(b)** **Standard Dosage –** Organosilane anti-stripping agents are to be added at standard dosage rate of 0.05% additive by weight of asphalt cement. All other anti-

stripping agents are to be added at a standard dosage rate of 0.3% additive by weight of asphalt cement.

**(c) Increased Dosage –** The anti-stripping agent dosage rate shall be increased as necessary to meet a minimum TSR value of 80.

**(d) Dosing Location –** The anti-stripping agent shall be added:

**(i)** at the AC supplier's terminal with the dosage rate percent included on the bill-of-lading; or

**(ii)** at the Contractor's asphalt mix plant only after the Ministry Representative has approved the process for blending the materials, conducting QC, and monitoring and confirming dosage rates.

**(e) Payment –** All costs associated with supplying and blending the additive into the asphalt cement shall be incidental and included in the Unit Price bid for the varying classes and lifts of asphalt mix.

If an increased dosage of anti-stripping agent is required above the standard dosage rates specified SS 502.08.03(b) above, or as amended in the Special Provisions, the costs for the additional anti-stripping agent above the standard dosage will be paid through the Provisional Sum for Site Modifications.

**502.08.04 Asphalt Mix Design Submittals –** The Contractor shall submit each Asphalt Mix Design to the Ministry Representative for review. The Ministry will review the Asphalt Mix Design to ensure it complies with the requirements of the Contract.

Paving Work shall not proceed until the Contractor receives the results of the review in writing from the Ministry Representative. The Contractor's submissions shall include the following information:

**(a)** Gradation (ASTM C117) of each aggregate to be used in each mixture;

**(b)** Percentage by mass of each aggregate to be used in each mixture;

**(c)** Design gradation (ASTM C117) of the combined aggregate for each mixture for each of the sieve sizes applicable to the mix, per the first column of Table 502-D;

**(d)** Estimated dry sieve gradation (ASTM C136) correlated to the JMF gradation;

**Table 502-D: Asphalt Mix Aggregate Gradation Limits**

| Sieve Size (mm) | Percentage Passing by Mass | | | | | | |
|---|---|---|---|---|---|---|---|
| | Coarse Mix | Medium Mix | | Fine Mix | Asphalt Base Course (ABC) | Asphalt Bound Open Graded Base (ABOGB) | Superpave |
| | 37.5 mm | 19 mm | 16 mm | 12.5 mm | 25 mm | 25 mm | Nominal 12.5 mm |
| 37.5 | 100 | | | | | | |
| 25.0 | 80 – 100 | | | | 100 | 100 | |
| 19.0 | 60 – 92 | 100 | | | 80 – 94 | 75 – 100 | 100 |
| 16.0 | | | 100 | | | | |
| 12.5 | 50 – 85 | 84 – 95 | 90 – 100 | 100 | | | 90 – 100 |
| 9.50 | 40 – 80 | 73 – 90 | 73 – 90 | 90 – 100 | 50 – 84 | 30 – 60 | |
| 4.75 | 30 – 65 | 50 – 75 | 50 – 75 | 55 – 80 | 25 – 55 | 5 – 30 | |
| 2.36 | 20 – 50 | 35 – 57 | 35 – 57 | 32 – 64 | 20 – 45 | 0 – 10 | 28 – 58 |
| 1.18 | 15 – 35 | 26 – 45 | 26 – 45 | 24 – 51 | 15 – 35 | | |
| 0.600 | 8 – 30 | 18 – 34 | 18 – 34 | 17 – 40 | | | |
| 0.300 | 6 – 22 | 10 – 26 | 10 – 26 | 13 – 29 | 5 – 20 | 0 – 8 | |
| 0.150 | 3 – 15 | 6 – 17 | 6 – 17 | 8 – 18 | | | |
| 0.075 | 1 – 7 | 3 – 7 | 3 – 7 | 4 – 10 | 0 - 5 | 0 - 4 | 2 – 10 |

**Table 502-E: Marshall Design and Production Criteria**

| Property of Laboratory Compacted Paving Mixture | Pavement Class | |
|---|---|---|
| | 1 | 2 |
| Number of blows each face of test specimens | 75 | 75 |
| Minimum % Voids in Asphalt Mix Aggregate for 19 mm Medium Asphalt Mix | 14 | 14 |
| Minimum % Voids in Asphalt Mix Aggregate for 16 mm Medium Asphalt Mix | 14.5 | 14 |
| Minimum % Voids in Asphalt Mix Aggregate for 12.5 mm Fine Asphalt Mix | 15 | 15 |
| % air voids in laboratory compacted mixture for 19 mm Medium Asphalt Mix | 2.5 to 4.5 | 2.5 to 4.5 |
| % air voids in laboratory compacted mixture for 16 mm Medium Asphalt Mix | 2.5 to 4.5 | 2.5 to 4.5 |
| % air voids in laboratory compacted mixture for 12.5 mm Fine Asphalt Mix | 2.5 to 4.5 | 2.5 to 4.5 |
| Minimum Marshall Load, N @ 60°C for 80 – 100 Pen. and 120 – 150 Pen. | 9000 | 7000 |
| Minimum Marshall Load, N @ 60°C for 150 – 200 Pen. and 200 – 300 Pen. | 7000 | 6000 |
| Flow index, units of 0.25 mm | 8 to 14 | 8 to 16 |
| Minimum Asphalt Film Thickness, $\mu$m (microns) *(see APPENDIX 502-E )* | 8.0 | 8.0 |
| Minimum Index of Retained Stability after immersion in water at 60°C for 24 hours | 85% | 75% |

**Table 505–F: Superpave Design and Production Criteria**

| **Property of Laboratory Compacted Paving Mixture For Design ESALs = 10 - 30 million, at $N_{Design}$ = 100 gyrations** | **SuperPave** |
|---|---|
| % voids in the Mineral Aggregate, minimum | 14 |
| Required density: % of Theoretical Maximum Specific Gravity in a laboratory compacted mix<br>at $N_{max}$ = 160 gyrations<br>at $N_{Design}$ = 100 gyrations<br>at $N_{Initial}$ = 8 gyrations | <br>98<br>95 – 97[1]<br><89 |
| % voids filled with Asphalt Cement | 65 – 75 |
| Dust to Binder ratio[2] | 0.6 – 1.2 |

[1] *For design Superpave density shall be 96*

[2] *Consideration shall be given to increasing the dust to binder ratio to 0.8 – 1.6*

**(e)** All Asphalt Mix Design data used in arriving at the final mix designs;

**(f)** Design Asphalt Content expressed as a percentage of the dry weight of the aggregate;

**(g)** Design Mix Temperature, which shall be the temperature at which the kinematic viscosity of the asphalt cement is 0.17 Pa•s (170 centistokes) or as per the Asphalt Cement supplier's recommendations; and

**(h)** Recommended compaction temperature.

**502.08.05 Ministry's Review of Asphalt Mix Design –** The Ministry Representative will require up to seven (7) days from the time of receipt of the Asphalt Mix Design for review. The mix design must be reviewed and accepted by the Ministry Representative prior to commencement of pavement construction.

Upon acceptance of the Asphalt Mix Design, the Contractor shall prepare the laboratory equipment calibration samples, submit them to the Ministry Representative, and participate in the calibration and correlation process described in APPENDIX 502-C Blank Aggregate Sample Preparation and APPENDIX 502-D Ignition Oven Correlation Procedure.

**502.08.06 Ignition Oven Correlation –** After receiving acceptance of the Asphalt Mix Design, the Contractor shall prepare blank aggregate samples (aggregate-only samples prepared to match the mix design) for correlation of the Contractor, Ministry and appeal laboratory ignition ovens. Blanks shall be prepared in accordance with APPENDIX 502-C Blank Aggregate Samples.

The Ministry Representative shall randomly select which of the individual blanks will be used.

Within five (5) days, and prior to any mix production, the Contractor and the Ministry shall prepare and test asphalt mix samples in accordance with APPENDIX 502-D Ignition Oven Correlation.

**502.08.07 Use of Calibration Factors in Reporting Asphalt Cement Content –** Results from testing of any asphalt mix shall report the measured AC Content and the corrected AC Content after applying the applicable laboratory calibration factor.

**502.08.08 Verification of Job Mix Formula from the Asphalt Mix Design –** Verification of the Asphalt Mix Design will be carried out by the Contractor during the course of production of the first 1,000 tonnes of mix using the Asphalt Mix Design.

During the first 1,000 tonnes of plant production, the Contractor may make any adjustments it chooses to the Asphalt Mix Design, testing the mix, and refining the Asphalt Mix Design to a state that fully complies with Table 502-D and Table 502-E (or Table 505–F as applicable) and the Special Provisions.

All mix of the Asphalt Mix Design laid must be tracked by the Contractor, and reported to the Ministry Representative, as to lay-down location and the Asphalt Mix Design values in effect at the time that mix was produced, to ensure appropriate values are used in comparing design to sampled properties.

After production of the first 1,000 tonnes, the Contractor shall declare their Job Mix Formula (JMF) to the Ministry Representative and provide volumetric properties/test data on the final mix produced. Any future adjustments to the JMF shall comply with all requirements of this SS 502.

Where the JMF varies from the Asphalt Mix Design by a cumulative amount greater than any tolerance specified in Table 502-G, the Contractor shall do a single point confirmatory Asphalt Mix Design and report the results to the Ministry Representative.

**502.08.09 AC Content Bump –** Upon receipt of a Job Mix Formula meeting all Contract requirements, the Ministry may direct the Contractor to increase the asphalt cement content by a "bump" of up to 0.3% by weight of dry aggregate in the mix.

**502.08.10 Field Adjustment of Job Mix Formula –** A field adjustment to the Job Mix Formula is defined as a change in the asphalt cement content of the mix, aggregate gradation and/or proportioning of various aggregate sizes, within the specified limits as shown Table 502-G without review and acceptance of a new Asphalt Mix Design.

The proposed field adjustment shall be submitted in writing together with supporting documentation to the Ministry Representative. Within four hours of receipt of the proposed field adjustments, the Ministry Representative will review the field adjustment for conformance with the contract requirements and notify the Contractor whether or not it is acceptable.

The Contractor's field adjustment to the Job Mix Formula must comply with the Asphalt Mix Design requirements of SS 502.08.02 through SS 502.08.04 inclusive. The Contractor shall provide all supporting verification data.

After the Job Mix Formula has been established in accordance with SS 502.08.10, no field adjustment to that Job Mix Formula will be permitted without prior written authorization by the Ministry Representative. The Ministry Representative will limit the number of field adjustments of the Job Mix Formula to two from the originally derived Asphalt Mix Design.

No field adjustment will be acceptable if it results in a change from the Asphalt Mix Design, for any property, in excess of the maximum adjustment for that property permitted in Table 502-G. All production Mix shall meet either Table 502-E or Table 505–F.

**Table 502-G: Field Adjustment of Job Mix Formula**

| Job Mix Formula Property | Maximum Cumulative Field Adjustment |
|---|---|
| **Percentage Passing by Sieve Size:** | |
| • 37.5 mm, 25.0 mm, 19.0 mm, and 16.0 mm | ±2.0% |
| • 12.5 mm and 9.5 mm | ±2.0% |
| • 4.75 mm and 2.36 mm | ±1.5% |
| • 1.18 mm and 0.600 mm | ±1.5% |
| • 0.300 mm and 0.150 mm | ±1.5% |
| • 0.075 mm | ±0.5% |
| **Asphalt Cement content** | ±0.3% |

## EQUIPMENT AND PLANT

**502.15** All equipment and plant shall be in good mechanical condition and be capable of performing the Work in accordance with this section.

## CONSTRUCTION

**502.20 Minimum Acceptable Construction Practices –** Professional standards in accordance with the Contractor's QC Plan and construction industry best practices are a core requirement of the Work. Any construction practice or activity that results in an obvious defect must be corrected